

Team Name

LEGAL VISIONARIES

Project Title

LegalEase – AI-Powered Legal Document Generator

Team Members

1. **Rithika**
2. **Sindhuja**
3. **Sumathi**
4. **Sathvika**

PHASE 4: PROJECT PLANNING

1. Introduction

The Project Planning phase defines the activities, responsibilities, milestones, and development schedule for the LegalEase project.

The project will be developed by dividing the work among the four team members according to their assigned responsibilities. The team will coordinate the different modules and integrate them into a single working application.

2. Project Goals

The main goals of the project are:

1. Develop a functional AI-powered legal document generator.
2. Create a simple and user-friendly frontend.
3. Develop the backend API for processing document requests.
4. Integrate Gemini AI for document generation.
5. Provide editable document previews.
6. Support TXT, DOCX, and PDF exports.
7. Test the complete application.
8. Prepare project documentation and demonstration materials.

3. Team Responsibility Allocation

Team Member	Role	Main Responsibilities
Rithika	Frontend / UI	Streamlit interface, input forms, preview interface, UI improvements
Sindhuja	Backend	FastAPI, API endpoints, request handling, backend integration
Sumathi	AI Integration	Gemini integration, prompts, AI response handling
Sathvika	Testing & Documentation	Test cases, bug tracking, documentation, final project records

4. Major Project Activities

The project will be completed through the following activities:

Activity 1 – Environment Setup

- Install Python and required packages.
- Set up the project directory.
- Create the initial project structure.
- Configure Git and GitHub.
- Configure environment variables.

Activity 2 – Frontend Development

- Create Streamlit application.
- Create document type selection.
- Create input fields.
- Create Generate Document button.
- Create document preview.
- Add editing and download controls.

Responsible: Rithika

Activity 3 – Backend Development

- Create FastAPI application.
- Create API routes.
- Create /generate endpoint.
- Validate incoming requests.
- Connect the backend with the AI module.

Responsible: Sindhuja

Activity 4 – AI Integration

- Configure Gemini API integration.
- Develop the AI generation module.
- Prepare prompts for different document types.
- Process AI-generated responses.
- Connect AI generation with the backend.

Responsible: Sumathi

Activity 5 – Document Formatting and Export

- Process generated document content.
- Implement TXT export.
- Implement DOCX export.
- Implement PDF export.
- Apply professional formatting.

Team Activity: Rithika, Sindhuja and Sumathi

Activity 6 – Testing

- Test different document types.
- Test valid and invalid inputs.
- Test AI-generated output.
- Test document preview.
- Test editing functionality.
- Test TXT, DOCX and PDF downloads.
- Record defects and corrections.

Responsible: Sathvika

Activity 7 – Documentation

- Maintain phase-wise documentation.
- Record development progress.
- Prepare test documentation.
- Prepare final project documentation.
- Organize project files.

Responsible: Sathvika

Activity 8 – Demonstration

- Prepare the final application.
- Prepare demonstration scenarios.
- Record the project demonstration.
- Prepare the final presentation materials.

Team Activity: All members

5. Project Milestones

Milestone Expected Outcome

M1	Brainstorming & Ideation completed
M2	Requirement Analysis completed
M3	Project Design completed
M4	Project Planning completed
M5	Basic application developed
M6	AI integration completed
M7	Document export and formatting completed
M8	Testing completed
M9	Documentation completed
M10	Demonstration completed

6. Development Schedule

The project will follow a sequential development plan.

Stage Planned Work

Stage 1 Project setup and environment configuration

Stage 2 Frontend development

Stage 3 Backend development

Stage 4 AI integration

Stage 5 Document formatting and export

Stage 6 Module integration

Stage 7 Testing and debugging

Stage 8 Documentation

Stage 9 Demonstration preparation

The exact duration of each activity may be adjusted according to the team's development progress and SmartBridge submission schedule.

7. Task Dependencies

The major dependencies are:

Project Setup



Frontend + Backend Development



AI Integration



Module Integration



Document Export



Testing



Documentation



Demonstration

The frontend and backend can be developed in parallel. AI integration can also be prepared independently before the modules are fully integrated.

8. Risk Planning

Risk	Possible Impact	Planned Action
API problems	AI generation may not work	Check API configuration and error handling
Invalid user input	Incorrect document generation	Validate inputs
AI-generated errors	Incorrect or incomplete content	Test different inputs and review output

Integration issues	Modules may not communicate correctly	Perform integration testing
Formatting problems	Poor document appearance	Test TXT, DOCX and PDF outputs
Development delays	Project schedule may be affected	Divide tasks clearly and track progress
GitHub issues	Difficulty managing project files	Use regular commits and organized folders

9. Team Coordination

The team will coordinate development activities by:

1. Dividing tasks according to responsibilities.
2. Sharing progress regularly.
3. Testing integrated modules together.
4. Recording problems and solutions.
5. Maintaining project files in an organized structure.
6. Using GitHub for project version control and collaboration.

10. Version Control Plan

The project will use Git and GitHub to maintain source-code versions.

The team will maintain:

- Source code
- Project documentation
- Requirements
- Testing records
- README file

API keys and other sensitive credentials should not be uploaded to the public repository.

11. Completion Criteria

The project will be considered ready for final demonstration when:

1. The frontend is functional.

2. The backend API is functional.
3. Gemini AI integration is working.
4. Legal document generation is working.
5. Generated documents can be previewed and edited.
6. TXT, DOCX, and PDF exports work correctly.
7. Major test cases have been completed.
8. Required documentation is prepared.
9. The final application can be demonstrated successfully.

12. Conclusion

The Project Planning phase establishes the responsibilities, activities, milestones, dependencies, risks, and development approach for LegalEase.

By dividing the work among the team members and coordinating the frontend, backend, AI integration, testing, and documentation activities, the team can systematically develop and demonstrate the LegalEase application.